

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester VI)
Artificial Intelligence Practical

Practical – IV(Task)

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Class : TYIT	Date/Time of Submission : 25/07/2026

Aim: Bayes' Rule Application

Q.1 In a college:

- 20% of students know Python.
- 80% of students know Java.
- 90% of Python students get placed.
- 40% of Java students get placed.

A randomly selected student is placed.

Find:

- Probability that the student knows Python.
- Probability that the student knows Java.
- Probability that a placed student knows Python.

Code:

Probability of knowing Python

P_Python = 0.20

Probability of knowing Java

P_Java = 0.80

Probability of placement if Python

P_Placed_Python = 0.90

Probability of placement if Java

P_Placed_Java = 0.40

Bayes Rule

```
P_Python_Placed = (P_Placed_Python * P_Python) / (
    (P_Placed_Python * P_Python) +
    (P_Placed_Java * P_Java)
)
```

```
print("Probability that the placed student knows Python:")
```

```
print(round(P_Python_Placed * 100,2),"%")
```

```
print("Divyanjali yadav T052")
```

Output:

```

0.05
Probability that the placed student knows Python:
36.0 %
Divyanjali yadav T052

```

Q.2 In a city:

- 85% of taxis are Green.
- 15% are Blue.
- A witness correctly identifies the taxi color 80% of the time.

A witness claims the taxi involved in an accident was Blue. What is the probability that the taxi was actually Blue?

Code:

```
# Probability of Blue taxi
```

```
P_Blue = 0.15
```

```
# Probability of Green taxi
```

```
P_Green = 0.85
```

```
# Witness says Blue if taxi is Blue
```

```
P_SaysBlue_Blue = 0.80
```

```
# Witness says Blue if taxi is Green
```

```
P_SaysBlue_Green = 0.20
```

```
# Bayes Rule
```

```
P_Blue_SaysBlue = (P_SaysBlue_Blue * P_Blue) / (
    (P_SaysBlue_Blue * P_Blue) +
    (P_SaysBlue_Green * P_Green)
)
```

```
print("Probability that the taxi was actually Blue:")
```

```
print(round(P_Blue_SaysBlue * 100,2),"%")
```

```
print("Divyanjali yadav T052")
```

Output:

```

Probability that the taxi was actually Blue:
41.38 %
Divyanjali yadav T052

```